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Improving T5 for mFDA-Aligned Clinical Reporting from Acoustic Biomarkers

Problem statement & Motivation

Clinical assessment of Parkinsonian dysarthria relies on subjective, time-consuming mFDA ratings. We generate mFDA-aligned reports from acoustic biomarkers computed on four tasks (/a/, /pa-ta-ka/, reading, conversation) in PC-GITA, mapping features to the seven mFDA categories to standardize reporting, reduce workload, and enable objective screening/monitoring; improving compact T5/Flan-T5 toward 7B performance supports privacy-preserving, cost-effective on-prem deployment.

Proposed approach

We aim to improve T5/Flan-T5 for generating structured, mFDA-aligned clinical speech reports directly from acoustic biomarkers, closing the gap to 7B models while remaining lightweight for privacy-preserving, on-prem deployment. Methods include: carefully designed prompt schemas that encode task, category headers, and normalized feature tokens; parameter-efficient fine-tuning (LoRA) on T5/Flan-T5; curriculum over severity bands; and higher-fidelity simulation of biomarker–report pairs that preserves per-task and per-category correlations. We will compare T5 variants and PEFT configurations, tune lr/weight-decay/batch size, and ensure full category coverage in the targets.

The dataset is PC-GITA (50 PD, 50 HC; 16 kHz; tasks: sustained /a/, /pa-ta-ka/, reading, conversation) with mFDA labels from three phoniatricians (median; 0/1 Normal, 2 Mild, 3 Moderate, 4 Severe). We compute task-specific biomarkers—phonation (F0 SD, jitter, shimmer), prosody (pause metrics, pitch/SPL variability), and phonemic precision (posterior probabilities, LLR, PVI)—map them to the seven mFDA categories, and tokenize them for input. Validation uses: (1) training on 110k simulated pairs with stratified severities; (2) evaluation on the 100 real reports held out for test using BLEU, plus per-category coverage checks; and (3) optional blinded clinician review for clinical plausibility. Baseline optimization follows 3 epochs, lr≈3e-4, weight decay≈0.1, batch size=8 on a single A100, with early stopping on a synthetic dev split.

Proposed Pipeline

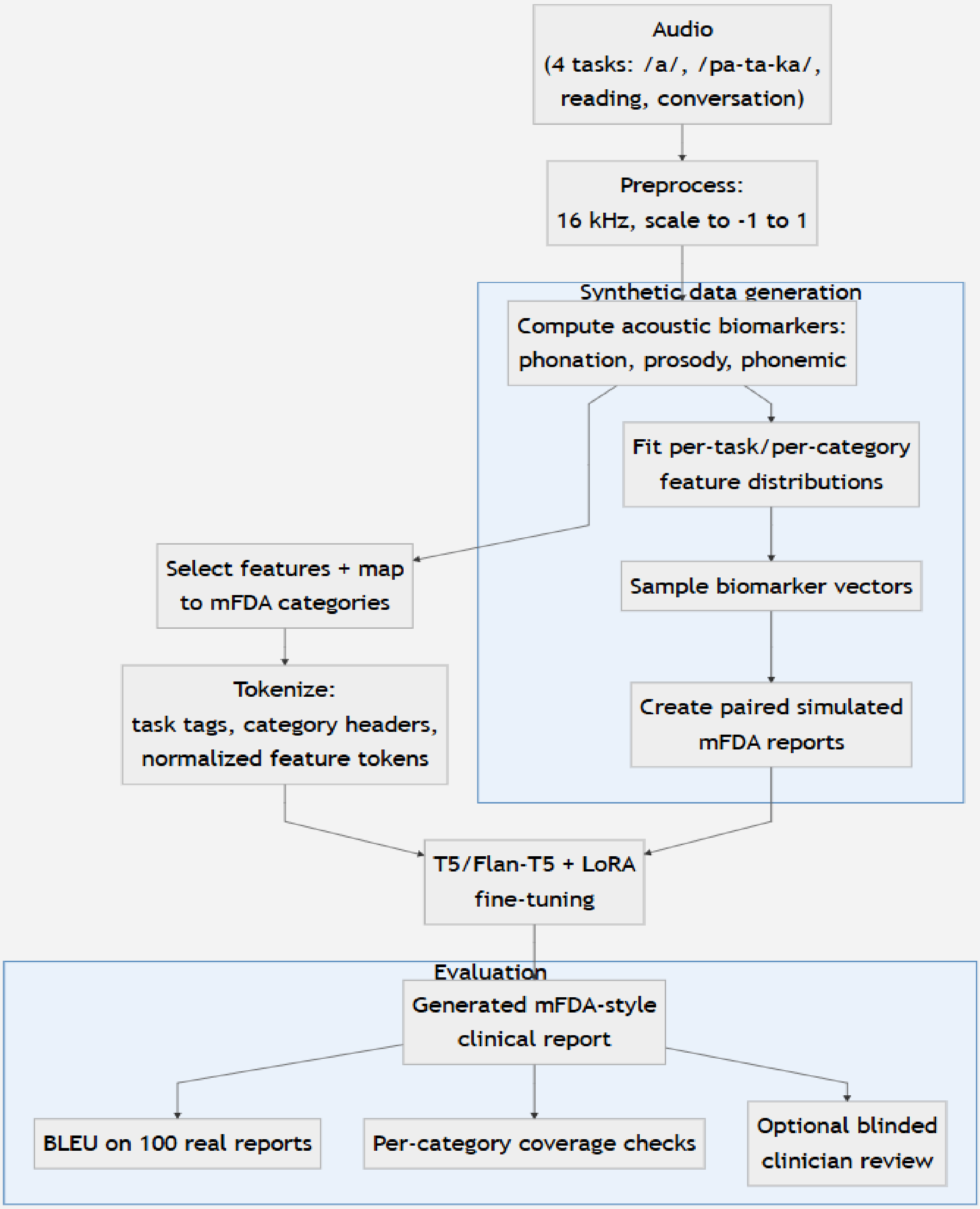

Figure: End-to-end pipeline from audio to tokenized biomarker prompts and fine-tuned T5/Flan-T5 outputs, with a synthetic data branch for scalable supervision and evaluation on held-out real reports.

Research question

Our research asks whether compact encoder-decoder LLMs (T5/Flan-T5) can learn a clinically faithful, interpretable mapping from task-specific acoustic biomarkers to structured mFDA-style reports, and under what training and augmentation conditions they approach 7B-model performance. We will study how prompt schema design, PEFT (LoRA), and higher-fidelity synthetic biomarker–report pairs affect accuracy, category coverage, and calibration across severity levels. Beyond BLEU, we will test generalization to unseen speakers/tasks, robustness to recording shifts, and alignment with clinician ratings via blinded review. The goal is to trade minimal compute for maximal clinical validity and reproducibility.

Objectives

General:

Improve T5/Flan-T5 to generate clinically faithful, mFDA-style speech reports from acoustic biomarkers with accuracy approaching 7B models while remaining compute-efficient.

Specific:

- Design and ablate prompt/tokenization schemas mapping task-specific biomarkers to the seven mFDA categories to maximize BLEU and per-report category coverage.
- Apply PEFT (LoRA) and targeted hyperparameter search on T5/Flan-T5 to close the BLEU gap to LLaMA-7B/Mistral-7B on the 100 real reports under latency/memory constraints.
- Develop higher-fidelity synthetic biomarker→report generation that preserves per-task/per-category correlations, improving generalization to unseen speakers/recording shifts and clinician plausibility.

Timeline

Month	Tasks / Milestones
November 2025	Literature review on pathological speech assessment and LLM-based medical report generation. Reproduce baseline results from the paper using existing models (T5, Flan-T5, LLaMA-7B).
December 2025	Design prompt schemas and implement LoRA fine-tuning on T5/Flan-T5. Generate and validate synthetic biomarker–report pairs with stratified severity.
January 2025	Conduct model training and hyperparameter tuning (learning rate, weight decay, batch size). Evaluate BLEU and per-category coverage on real test reports. Begin blinded clinician review for qualitative validation.
Febraury 2025	Analyze results, prepare comparative tables against 7B models, and write final report. Submit code, documentation, and final thesis deliverables.